

Quiz 2 Name Brandon Buchanan

1)

- a) How many digits to represent 102,103 in base 10 (Hint for next problem)? $\log_{10} m$
- b) How many digits to represent M in base 10? $0 < \frac{m}{10^x} < 10 - \lfloor \log m \rfloor$
- c) How many digits to represent M in base 2? $0 < \frac{m}{2^x} < 2 - \lfloor \log m \rfloor$

2) Suppose you have n distinct numbers, $x_1, x_2, \dots, x_n$ and that you are sorting the numbers

- a) How many different possible outcomes are possible for the rearranged sorted numbers

$n!$

For example if you have two numbers x_1 and x_2 there are two different arrangements that the final sorting could produce x_1, x_2 or x_2, x_1

1 2 3
1 3 2
2 1 3
2 3 1
3 2 1
3 1 2

3) Write code for binary search and find running time

```
BinarySearch(int[] array, int valueToFind, int low, int high)
{
    int halfway = (high - low) / 2 + low;
    if (array[highway] > valueToFind)
    {
        return BinarySearch(array, valueToFind, low, array[highway]);
    }
    else if (array[highway] < valueToFind)
    {
        return BinarySearch(array, valueToFind, array[highway], high);
    }
    else
        return array[highway];
}
```

1 2 3 4
1 2 4 3
1 3 2 4
1 3 4 2
1 4 3 2
1 4 2 3

4)

Suppose you have a binary tree with n nodes

- a) What is the minimum depth of the tree? 0
- b) What is the maximum depth of the tree? $\lg n$

5)

Multiplication two N bit numbers

Write code to multiply these two numbers

$A[1] \dots A[N]$

$B[1] \dots B[n]$

Write code for number multiplication and find running time.